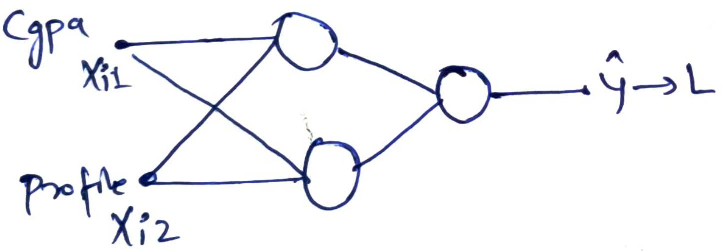
→ Back propagation → The why?
The intuition behind the algorithm.
Back propagation Algorithm

for i in range (epochs):
    for j in range (X, shape[0]):
        Select 1 row (random)
        Predict (using Forward prop)
        Calculate loss (using loss function → mse)
        Update weights and bias using GD
            $W_n = W_o - \eta \frac{\partial L}{\partial W}$
    Calculate avg loss for the epoch

→ Loss function is a function of all trainable parameters

$L = (y - \hat{y})^2 \qquad L(\hat{y})$

y ↙ constant



$\hat{y} = W_{11}^2 O_{11} + W_{21}^2 O_{12} + b_{21}$

$\hat{y} = W_{11}^2 [W_{11}^1 X_{i1} + W_{21}^1 X_{i2} + b_{11}] +$
$W_{21}^2 [W_{12}^1 X_{i1} + W_{22}^1 X_{i2} + b_{12}] + b_{21}$

$X_{i1}, X_{i2}, X_{i1}, X_{i2}$ → It is a constant

$L(w, b)$ → function of 9 thing

If we change any thing then L will be change.

→ Concept of Gradient:

↳ fancy word of derivative.

Complex $L(W^1_{11}, W^1_{12}, \ldots W^2_{21}, b_{21}, b_{11}, b_{12})$

gradient → $\frac{\partial L}{\partial W}$, $\frac{\partial L}{\partial b}$.

9 paramiter

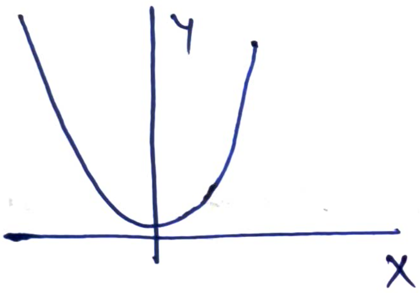
ek se jyada paramiter hota hai then hum gradient nikalte hai.

→ Concept of Derivative:

→ x mai change hone mai y mai ~~kay~~ kya ho rha hai.

→ Derivative at any point → slope at ~~any~~ that point.

→ Concept of Minima



$y = x^2$

$\frac{\partial y}{dx} = 2x = 0$

$\boxed{x = 0}$ minima at $x = 0$

$L(9 \text{ paramiter})$ $L\downarrow$

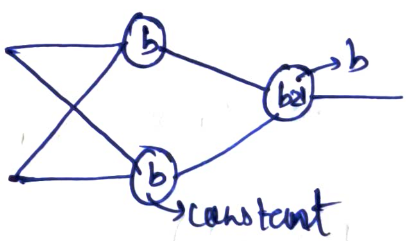
$\frac{\partial L}{\partial W^1_{11}} \text{ ---- } \frac{\partial L}{\partial b_{12}} = 0$

→ Backprop Intuition

$W_{new} = W_{old} - \eta \frac{\partial L}{\partial W}$

let $\eta = 1$

$W_{new} = W_{old} - \frac{\partial L}{\partial W}$



$L = L(w, b)$ → 9 ↘ 8 are constent assume

b_{21}

$L(b_{21})$ → only variable

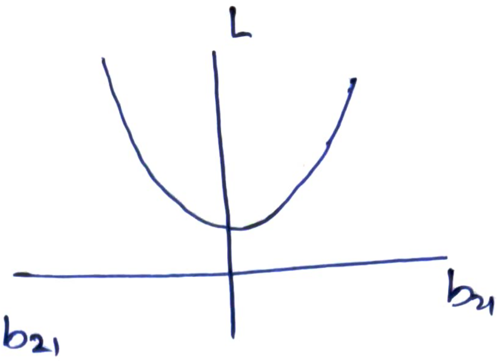
$L = (y - \hat{y})^2$

↳ $\hat{y} = (b_{21})$

$L(b_{21})$

$$b_{21} = b_{21} - \frac{\partial L}{\partial b_{21}}$$

$$L = b_{21}^2$$



$\frac{\partial L}{\partial b_{21}} \rightarrow$ derivative of L w.r.t b_{21}

To reduce L↓

$\frac{\partial L}{\partial b_{21}} \Rightarrow +ve \rightarrow b_{21}\uparrow \; L\uparrow$ (then we decrease b_{21} to decrease L)

$\frac{\partial L}{\partial b_{21}} = -ve \rightarrow b_{21}\uparrow \; L\downarrow$ (then we increase b_{21} to reduce L)

Effect of learning rate (η)

It make the step ~~smooth m~~ smooth.

What is Convergence

$$W_n = W_0 - \eta \frac{\partial L}{\partial W} = 0$$

$\boxed{W_n \approx W_0} \rightarrow$ It means ~~we~~ we reach minima.